

PREPARING FOR INFLUENZA SEASON

1. PROJECT OVERVIEW

This project is intended to help a medical staffing agency that provides temporary workers to clinics and hospitals on an as-needed basis. The analysis will help plan for the upcoming influenza season, a time when additional staff are in high demand.

MOTIVATION

The United States has an influenza season where more people than usual suffer from the flu. Some people, particularly those in vulnerable populations, develop serious complications and end up in the hospital. Hospitals and clinics need additional staff to adequately treat these extra patients. The medical staffing agency provides this temporary staff.

OBJECTIVE

Determine when to send staff, and how many, to each state.

SCOPE

The agency covers all hospitals in each of the 50 states of the United States, and the project will plan for the upcoming influenza season.

2. PROJECT MANAGEMENT PLAN

Stakeholder communication

- I. Stakeholder
 - Medical agency frontline staff (nurses, physician assistants, and doctors)
 - Hospitals and clinics using the staffing agency's service
 - Influenza patients
 - Staffing agency administrators

- II. Communication plan

Meetings:

1. At the project outset, there will be a meeting with the staffing agency administrators and representatives of hospitals and clinics using the staffing

agency's service. During this meeting, business requirements would be discussed.

2. Meetings will take place every 2 weeks with the staffing agency administrators and representatives of hospitals and clinics using the staffing agency's service to present and discuss progress of the project.
3. A last meeting will take place to present the results of the project and discuss the suggested staffing plan.

Calls:

Calls will be made to the different hospitals and clinics using the staffing agency's service to gather information on the specific staffing needs when needed.

Written Communication:

1. Emails will be sent to medical agency frontline staff and hospitals and clinics using the staffing agency's service to keep them informed about the staffing plan for the flu season. If there are changes in the staffing plan, these will be notified also via email on time.
2. Newsletters will be sent every 2 weeks to workers at hospitals and clinics using the staffing agency's service as well as to medical agency frontline staff to keep them updated on the progress of the project.
3. Advertisement will be created to approach to possible Flu patients to inform them about flu vaccination and other measures to take to reduce the possibility of infection.

Emergency/Contingency Plan:

For urgent matters, a meeting will be scheduled within the next two days and will be communicated by email requesting immediate confirmation.

Schedule and Milestones

[illegible]

Project Deliverables

- Written report with project requirements, objectives and scope. (Week 1)
- Written report consolidating the analytical insights. (Week 4)
- PPT with the results of the research project. (Week 9)
- Tableau Storyboard of the analysis. (Week 9)
- Video presentation for Stakeholders. (Week 9)

Audience Definition

1. Staffing agency administrators

The audience are experienced professional people directly related to the project. One can assume that they have a specific knowledge of the issue. They are in a management level, so they want concrete information about the progress of the analysis project.

2. Medical agency frontline staff (nurses, physician assistants, and doctors)

The audience may have a general knowledge of the issue. They work in the field but not in the management area. One can assume that they are not familiarized with statistics and numbers related to the issue, so they will need a more detailed explanation of the issue.

3. Hospitals and clinics using the staffing agency's service

This audience is divided into the representatives of the hospitals and clinics (which are in a management level) and the general workers of the hospitals (nurses, physician assistants, and doctors). So one should consider both audiences characteristics.

4. Influenza patients

The audience have a very basic knowledge of the issue (maybe because of the news about the hospitalizations in past flu seasons). One should consider that they want clear and understandable information in a colloquial language.

3. Business requirements translated into questions

1. When is the flu season?

- In temperate climates: mainly during winter.
- In tropical regions: throughout the year.

a. What determines the start and end of the season - number of people infected, specific time of the year, specific weather conditions?

As it seems, the weather conditions have an influence in the flu season.

b. Are there differences in the timing of flu seasons across states?

Yes, it depends on the climate in each state.

c. How have been the staff requirements in past flu seasons?

An important issue related to the staff during flu season is the absenteeism rate of hospitals and clinics staff.

2. Which flu cases result in hospitalizations?

Hospitalizations and death due to influenza occur mainly among high-risk groups.

a. Are there specific symptoms or risks to consider for hospitalization?

Pneumonia and sepsis. People with other medical issues.

b. Are vulnerable population more likely to be hospitalized due to the flu?

Yes, pregnant women, children under 5 years of age, older people (65 years and older), individuals with chronic illness (such as chronic cardiac, pulmonary, renal, metabolic, neurodevelopmental, liver or hematologic diseases) and individuals with immunosuppressive conditions/treatments (such as HIV, receiving chemotherapy or steroids, or malignancy) are more at risk than others. But also health and care workers due to increased exposure to the patients, and of further spreading particularly to vulnerable individuals.

4. HYPOTHESES

Hypothesis 1: If a person in a vulnerable population is infected by the flu, then the risk of dying from the flu increases compared to the general population.

Hypothesis 2: If a state has lower rates of flu vaccination, then it will have higher rates of hospitalization during the flu season compared to states with higher vaccination rates.

Hypothesis 3: if the flu primarily occurs during winter months, then the highest death rates among vulnerable populations will peak during this period.

Hypothesis 4: If a person is 65 or older and is not vaccinated against the flu, then the risk of death due to the flu is higher than that of younger individuals (18-64 years old).

Hypothesis 5: If the size of the vulnerable population (patient is older than. 65 years old) increases, then the risk of mortality increases.

5. DATA WISHLIST

- Flu mortality rate (grouped by age)
- Population demographics (age and population size)
- Vaccination rate (grouped by age)
- Flu mortality rate in people vaccinated (grouped by age)

Sources Consulted:

- World Health Organization. (n.d.). *Influenza (seasonal)*. Retrieved January 8, 2025, from [https://www.who.int/news-room/fact-sheets/detail/influenza-\(seasonal\)](https://www.who.int/news-room/fact-sheets/detail/influenza-(seasonal))
- Hampson, A. W. (2013). The influenza viruses. *Australian and New Zealand Journal of Public Health*, 37(2), 95-96. <https://doi.org/10.1111/1753-6405.12170>